

AR control unit

Safety instructions and other important information



Installation and operating instructions

<http://net.grundfos.com/qr/i/91834764>

English (GB) Safety instructions

Original safety instructions

These safety instructions give a quick overview of the safety precautions to be taken in connection with any work on this product.

Observe these safety instructions during handling, installation, operation, maintenance, service and repair of this product.

These safety instructions is a supplementary document, and all safety instructions will appear again in the relevant sections of the installation and operating instructions.

Keep these safety instructions at the installation site for future reference.

Information about the product

Warning



The enclosure class is only met if the sockets are protected! The data regarding the enclosure class applies to AR control units with correctly inserted plugs or screwed-on caps.



The AR control unit is NOT approved for operation in potentially explosive areas!

Caution

Pumps with AR control unit are only suitable for indoor use! Do not install outdoors!

Caution

Without Hall sensor the linear deviation will deteriorate to more than 10 %!

With Hall sensor, the linear deviation will be less than 2 %.

Installation



Warning

Observe the specifications for the range of applications and installation location described in sections [2. General information](#) and [4.2 Installation location](#).

Carefully tighten the screws. Otherwise the plastic housing may be damaged.

The function of the finally mounted control unit must be tested together with the dosing pump!

Caution

The nameplate must be stuck to the mounting plate or to the upper part of the control unit!

The installation site must be under cover! Ensure that the enclosure class of motor and pump is not affected by the atmospheric conditions.

Caution

Pumps with AR control unit are only suitable for indoor use! Do not install outdoors!

Electrical connections

Warning

Electrical connections must only be carried out by qualified personnel!

Disconnect the power supply before connecting the power supply cable and the relay contacts!



Before connecting the power supply cable, check that the rated voltage stated on the pump nameplate corresponds to the local conditions!

Do not make any changes to the power supply cable or plug!

Observe the local safety regulations!



Warning

The pump housing must only be opened by personnel authorised by Grundfos!



Warning

Protect the cable connections and plugs against corrosion and humidity.

Only remove the protective caps from the sockets that are being used.

Caution

The power supply must be electrically isolated from the signal inputs and outputs.

Caution

The pump can be automatically started by connecting the power supply!

Caution

Before switching on the pump, check that it is installed correctly. See section [4. Installation](#).

Mounting

The control unit is factory-mounted on the pump terminal box by Grundfos!

Caution

At each intervention in the control unit, the e.s.d. (electrostatic discharge) protection must be observed!

Connecting signal lines for AR control unit

Warning

Electrical connections must only be carried out by qualified personnel!



Disconnect the power supply before connecting the power supply cable and the relay contacts!

Observe the local safety regulations!



Warning

The pump housing must only be opened by personnel authorised by Grundfos!

**Warning**

Protect the cable connections and plugs against corrosion and humidity.

Only remove the protective caps from the sockets that are being used.

**Warning**

The enclosure class is only met if the sockets are protected! The data regarding the enclosure class applies to AR control units with correctly inserted plugs or screwed-on caps.

Caution

The power supply must be electrically isolated from the signal inputs and outputs.

There is a risk that the unit connected to the current output could be destroyed!

If the pump connected to the current output has a GND-PE reference, the current output may increase to a multiple of 20 mA!

Caution

If this is the case, connect an isolating amplifier between the current output and the unit.

**Warning**

For the connection of one cable, use a plug adapter with simple cable entry, for the connection of two cables, use a plug adapter with double cable entry, otherwise the protection will be lost!

Caution

The current output is not designed for controlling secondary pumps in master/slave operation.

Caution

For the connection of one cable, use a plug adapter with simple cable entry, for the connection of two cables, use a plug adapter with double cable entry, otherwise the protection will be lost!

Using the pump electronics

Caution

The motor monitoring function does not provide protection for the motor as power is still supplied to the motor. Therefore, the motor should be protected by a motor protector.

Caution

There is a risk that the connected evaluation unit could be destroyed! If the unit connected to the current output has a GND-PE reference, the current output may increase to a multiple of 20 mA!

If this is the case, connect an isolating amplifier between the current output and the unit.

Risk of damage to secondary pump!

The current signal is not electrically isolated and can damage slave pumps.

Do not control slave pumps with the current signal. Only control them with the electrically isolated contact input signal!

Caution

Risk of malfunction or damage to the pump!

Do not throw or drop the pump.

Caution

Unused contact inputs from the master pump are forwarded to the slave pumps. They are processed on the slave pumps according to the slave pump settings!

Caution

Subject to alterations.

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